Principle of the Excluded Middle

A statement is either true or false.

There is no in between.

There are either ∞ly many primes or there aren't
(No in between)

Well-Ordering Principle

Every non-empty subset of natural numbers has a least element.

Theorem. (Existence portion of Fundamental Theorem of Arithmetic)

Number Theory

Every natural number greater than 1 has a prime factorization.

This is a statement about infinitely many numbers

<u>Proof.</u> We will proceed by contradiction.

Suppose the theorem statement is false.
That is, suppose there exists at least one natural number greater than 1 without a prime factorization.
By the well-ordering principal we can select the smallest natural number >1 without a prime factorization.
Call it i. If i doesn't have any divisors d with $1<d<i$, then the only divisors of i are 1 and itself.
Thus i is prime, and $i=i$ is a prime factorization.
This is impossible, thus i has a divisor d with $1<d<i$. In other words, $i=d\cdot c$ with $1<d,c<i$.
Since $d,c<i$, they have prime factorizations:

$$d=p_1\cdots p_r \quad \text{and} \quad c=q_1\cdots q_s.$$

Therefore, $i=d\cdot c=p_1\cdots p_r\cdot q_1\cdots q_s$.
But this is a prime factorization of i!!!
Thus every natural number has a prime factorization. $\square$

Ex. $i = 7$. 7 is the smallest natural number without a prime factorization.

None of $2, 3, 4, 5, 6$ divide 7. Thus the only divisors of 7 are itself and 1.

Thus 7 is prime, and $7 = 7$ is a prime factorization.

Ex. $i = 24$

24 is the smallest natural number without a prime factorization.

24 is not prime if has divisors other than 1 & 24

$d = 4$ and $c = 6$.

$24 = 4 \cdot 6$ since $4, 6 < 24$ they have prime factorizations

$4 = 2 \cdot 2$, $6 = 2 \cdot 3$.

$24 = 4 \cdot 6 = 2 \cdot 2 \cdot 2 \cdot 3$.

This is a prime factorization of 24.

This shows 24 isn't the smallest number w/ out a prime factorization.

Theorem. There are ∞ly many prime numbers.

<u>Proof.</u> Suppose there are only finitely many prime numbers. I can list them all

$\overset{2}{p_1}, \overset{3}{p_2}, \overset{5}{p_3}, \overset{7}{\dots}, \overset{11}{p_n}$.

Let $Q = p_1 \cdot p_2 \cdot p_3 \cdot \cdots \cdot p_n = \prod_{i=1}^{n} p_i$.

Consider $Q+1$. We just proved $Q+1$ has a prime factorization.

In particular, let z be a prime dividing $Q+1$. Since we listed all the primes z is in $\{p_1, p_2, \ldots, p_n\}$. Thus z divides Q.

$z \mid Q$.

But $z \mid Q+1$. Thus $z \mid (Q+1-Q)$.

$$z \mid 1.$$

This is a contradiction.

The only divisor of 1 is 1.

and 1 is not prime.

Thus there are infinitely many primes. $\square$